# Computer Organization and Architecture (CSL251)
# LAB-8
# 2025-26 W

## 1 Unraveling the Secrets of Supervisor Mode:

We started our journey in the all powerful machine mode, but wait, there seems to be a hiccup. You need some help from the *Supervisor*. ***Switch from machine mode to supervisor mode*** and the path just might reveal itself.

## 2 Crafting Page Tables:

Behold! The time has come to weave the fabric of reality itself — page tables! With the wisdom of ancient scribes, design and forge the intricate page tables that shall guide your path through the labyrinth of virtual memory. All you need is the (hopefully not forgotten) advises from the *wise man* during class and your own intellect and will power to give you the power to ***Initialize the page tables at their right locations and set appropriate values. These are*** **Sv39-*pages***. Fight on warrior, we are almost there!

## 3 Unleashing the Power of Virtualization:

Harness the arcane forces of virtualization to transcend the limitations of physical reality. Channel your energies into ***Configuring the mystical control registers*** and awaken the dormant magic of virtual memory translation.

## 4 The Final Conquest: User Mode Awaits:

At long last, the ultimate prize beckons—user mode! With courage in your heart and code in your hand, take the leap of faith into the realm of users. Ensure that the gates are open wide, welcoming all who dare to tread the path you have paved. ***Let us go to the user mode!***

## 5 Test Your Mettle:

With your adventure complete, embark on one final quest: ***The validation of your accomplishments!*** Set forth into the virtual landscape, test every step of your journey, and ensure that the magic flows seamlessly through the veins of your creation.

To those amateur explorers who do not want to waste their precious time on the explanation riddled with smoke and mirrors (and some beauty of literature), here is a much straight forward ***"Beginner's Guide to Virtualization in RISCV (Vol. 101)":***

- ***Step-1:*** Switch execution from Machine Mode to Supervisor Mode.
    - ***Step-2:*** Initialise the page table, i.e. fill in the page table entries at appropriate the memory locations (refer to Fig.1). Keep in mind, we are doing *Sv39* -paging.

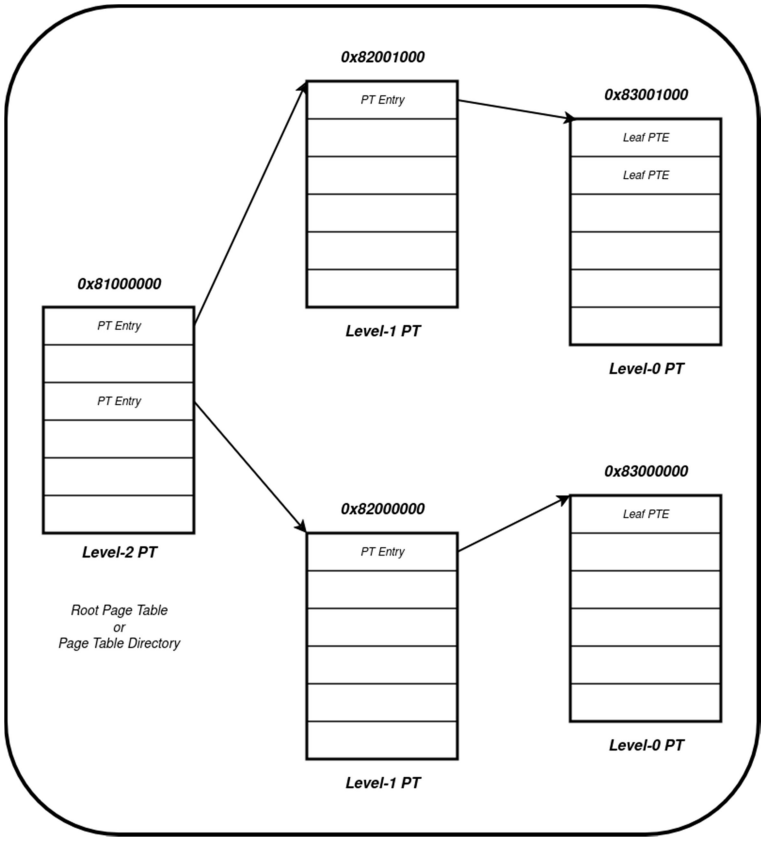



Figure 1: Page Table Structure for Sv39

- ***Step-3:*** Set the appropriate value at to the ***satp*** register by assigning a value to ***satp config***
- ***Step-4:*** Switch execution from Supervisor Mode to Machine Mode.
    - ***Step-5:*** Do you see that your physical addresses (the ones see in your *dump*-file) was replaced with their virtual addresses (when executing in spike)?

**** Keep in mind to align all the different components properly ****

- In case you forgot what your aim is, refer Fig.2.
    - In case you forgot what these registers do, how does the page translation occur, and what values do you set to the registers, go to *References.*

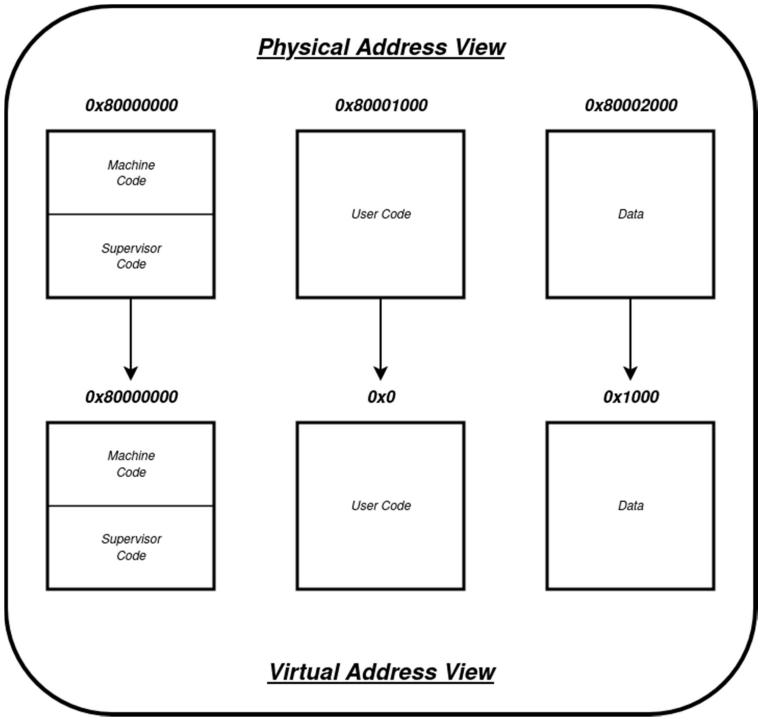



Figure 2: Pictorial view of the aim of the assignment

- Where to start? - The self-evident code template (va_template.S)
- How to switch between privilege modes? - Refer to your own code from the previous labs
- How to compile and create a dump file?

```
$ riscv64-unknown-elf-gcc -nostartfiles -T linker.ld va_template.S
$ riscv64-unknown-elf-objdump -D a.out > dump
```

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